

Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

Supplement to: Chi KN, Agarwal N, Bjartell A, et al. Apalutamide for metastatic, castration-sensitive prostate cancer. *N Engl J Med* 2019;381:13-24. DOI: 10.1056/NEJMoa1903307

Supplementary Materials for

Apalutamide in Metastatic Castration-Sensitive Prostate Cancer

Kim N. Chi, Neeraj Agarwal, Anders Bjartell, Byung Ha Chung, Andrea J. Pereira de Santana Gomes, Robert Given, Álvaro Juárez Soto, Axel S. Merseburger, Mustafa Özgüroğlu, Hirotugu Uemura, Dingwei Ye, Kris Deprince, Vahid Naini, Jinhui Li, Shinta Cheng, Margaret K. Yu, Ke Zhang, Julie S. Larsen, Sharon McCarthy, and Simon Chowdhury on behalf of TITAN investigators*

From BC Cancer and Vancouver Prostate Centre, Vancouver, BC, Canada (K.N.C.); Huntsman Cancer Institute, University of Utah, Salt Lake City, UT (N.A.); Skåne University Hospital, Lund University, Malmö, Sweden (A.B.); Yonsei University College of Medicine and Gangnam Severance Hospital, Seoul, South Korea (B.H.C.); Liga Norte Riograndense Contra O Cancer, Natal, Brazil (A.J.P.S.G.); Urology of Virginia, Eastern Virginia Medical School, Norfolk, VA (R.G.); Hospital Universitario de Jerez de la Frontera, Cádiz, Spain (A.J.S.); University Hospital Schleswig-Holstein, Campus Lübeck, Lübeck, Germany (A.S.M.); Istanbul University-Cerrahpaşa, Cerrahpaşa School of Medicine, Istanbul, Turkey (M.O.); Kindai University Hospital Faculty of Medicine, Osaka, Japan (H.U.); Fudan University Shanghai Cancer Center, Shanghai, China (D.Y.); Janssen Research & Development, Beerse, Belgium (K.D.); Janssen Research & Development, San Diego, CA (V.N., J.L., K.Z.); Janssen Research & Development, Raritan, NJ (S.Cheng, S.M.); Janssen Research & Development, Los Angeles, CA (M.K.Y., J.S.L.); Guy's, King's, and St Thomas' Hospitals, and Sarah Cannon Research Institute, London, UK (S. Chowdhury).

Contents

List of TITAN Investigators.....	3
SUPPLEMENTARY METHODS	9
SUPPLEMENTARY RESULTS	12
Supplementary Figure S1. CONSORT Diagram.....	13
Supplementary Figure S2. Kaplan-Meier Estimate of Time to Cytotoxic Chemotherapy.	14
Supplementary Figure S3. Kaplan-Meier Estimate of Time to PSA Progression.	15
Supplementary Figure S4. Kaplan-Meier Estimate of Time to Second Progression-free Survival.	16
Supplementary Figure S5. Repeated-Measures Mixed Effects Analyses for Mean Change from Baseline in FACT-P Total Score.....	17
Supplementary Table S1. Demographics and Baseline Disease Characteristics, Expanded List.	18
Supplementary Table S2. Prior Prostate Cancer Therapy.....	20
Supplementary Table S3. First Subsequent Systemic Prostate Cancer Therapy.....	21
Supplementary Table S4. Treatment-Emergent Adverse Events Leading to Treatment Discontinuation, Dose Reduction, and Dose Interruption.....	23
Supplementary Table S5. Investigator-Reported Cause of Death on Study in the Safety Population.....	33
Supplementary Table S6. Estimates of Subdistribution Hazard Ratio of the Apalutamide Group Relative to the Placebo Group.	35
SUPPLEMENTARY REFERENCES.....	36

List of TITAN Investigators.

Principal Investigator	Country
Batagelj, Emilio	Argentina
Bayo, Sergio	Argentina
Bella, Santiago	Argentina
De Romedi, Mario Luis	Argentina
Fein, Luis	Argentina
Greco, Martin Alberto	Argentina
Kahl, Susana	Argentina
Korbenfeld, Ernesto	Argentina
Montes de Oca, Luis Fernando	Argentina
Richardet, Martin	Argentina
Salvatierra, Alejandro	Argentina
Staneloni, Esteban	Argentina
Varela, Mirta	Argentina
Zarba, Juan Jose	Argentina
Begbie, Stephen	Australia
Hocking, Christopher	Australia
Krieger, Laurence	Australia
Steer, Christopher	Australia
Carcano, Flávio	Brazil
Coelho de Andrade, Carlos José	Brazil
Crippa, Alexandre	Brazil
Cruz, Felipe	Brazil
Damião, Ronaldo	Brazil
Delgado, Gilson	Brazil
Dettino, Aldo	Brazil
Faccio, Adilson	Brazil
Franke, Fabio	Brazil
Freitas, Marcelo	Brazil
Gomes, Andrea Juliana	Brazil
Marinho, Gisele	Brazil
Martins, Suelen	Brazil
Nahas, William	Brazil
Odebrecht, Roberto	Brazil
Panhoca, Renato	Brazil
Paula, Adriano	Brazil
Smaletz, Oren	Brazil
van Eyll, Brigitte	Brazil
Chi, Kim	Canada
Fleshner, Neil	Canada
Fradet, Yves	Canada
Shayegan, Bobby	Canada
Siemens, Robert	Canada
Gotto, Geoffrey	Canada
Chen, Zhiwen	China
Du, Chuanjun	China
Guo, Hongqian	China

He, Dalin	China
Hu, Qiang	China
Huang, Jian	China
Jin, Jie	China
Luo, Hong	China
Ma, Lulin	China
Pu, Jinxian	China
Qiu, Mingxing	China
Sun, Zhongquan	China
Tian, Ye	China
Wan, Ben	China
Xia, Shujie	China
Xu, Enci	China
Yang, Yong	China
Ye, Dingwei	China
Ye, Zhangqun	China
Zou, Qing	China
Andel, Ivan	Czech Republic
Brodak, Milos	Czech Republic
Capoun, Otakar	Czech Republic
Dvorak, Jan	Czech Republic
Holeckova, Petra	Czech Republic
Jarolim, Ladislav	Czech Republic
Matouskova, Michaela	Czech Republic
Mecl, Jan	Czech Republic
Perkova, Hana	Czech Republic
Stanek, Roman	Czech Republic
Student, Vladimir	Czech Republic
Vanasek, Jaroslav	Czech Republic
Zachoval, Roman	Czech Republic
Bennamoun, Mostefa	France
Guy, Laurent	France
Lebret, Thierry	France
Ruffion, Alain	France
Spaeth, Dominique	France
Tazi, Youssef	France
Timsit, Marc-Olivier	France
Eckert, Ralf	Germany
Feyerabend, Susan	Germany
Hammerer, Peter	Germany
May, Matthias	Germany
Merseburger, Axel	Germany
Suttman, Henrik	Germany
Szymula, Silvio	Germany
von Klot, Christoph	Germany
Bodoky, Gyorgy	Hungary
Géczi, Lajos	Hungary
Kulcsár, Dániel	Hungary
Nyirády, Péter	Hungary

Torzsok, Ferenc	Hungary
Trautmann, Tibor	Hungary
Kejzman, Daniel	Israel
Mermershtain, Wilmosh	Israel
Peer, Avivit	Israel
Rosenbaum, Eli	Israel
Sella, Avishay	Israel
Caserta, Claudia	Italy
Galli, Luca	Italy
Hamzaj, Afete	Italy
Milella, Michele	Italy
Morelli, Franco	Italy
Mosca, Alessandra	Italy
Mucciarini, Claudia	Italy
Procopio, Giuseppe	Italy
Scagliotti, Giorgio	Italy
Sciarra, Alessandro	Italy
Arai, Gaku	Japan
Fukasawa, Satoshi	Japan
Hashine, Katsuyoshi	Japan
Iijima, Kazuyoshi	Japan
Matsumoto, Ryuji	Japan
Nishimura, Kazuo	Japan
Nomura, Takeo	Japan
Sakai, Hideki	Japan
Suzuki, Hiroyoshi	Japan
Tsukino, Hiromasa	Japan
Tsumura, Hideyasu	Japan
Uemura, Hiroji	Japan
Uemura, Hirotsugu	Japan
Yamaguchi, Akito	Japan
Calvo, David	Mexico
Castro, Armando	Mexico
Rodriguez, Arturo	Mexico
Hernandez, Narciso	Mexico
Kuri, Roberto	Mexico
Martinez, Ivan	Mexico
Murillo, Emilio	Mexico
Garcia, Lizbett	Mexico
Humanski, Piotr	Poland
Jarzemski, Jacek	Poland
Marcheluk, Adam	Poland
Oszukowska, Elzbieta	Poland
Skoneczna, Iwona	Poland
Zaremba, Marcin	Poland
Byun, Seoksoo	Republic of Korea
Chung, Byungha	Republic of Korea
Hong, Sung Joon	Republic of Korea
Kang, SungGu	Republic of Korea

Kim, Choung Soo	Republic of Korea
Kwak, Cheol	Republic of Korea
Kwon, DongDeuk	Republic of Korea
Kwon, TaeGyun	Republic of Korea
Lee, Ji Youl	Republic of Korea
Lee, Kanghyun	Republic of Korea
Seo, SeongIl	Republic of Korea
Song, Ki Hak	Republic of Korea
Jinga, Viorel	Romania
Lungulescu, Dan	Romania
Militaru, Claudia	Romania
Alekseev, Boris	Russia
Budnik, Nikolay	Russia
Chistyakov, Valery	Russia
Gafanov, Rustem	Russia
Karyakin, Oleg	Russia
Khvorostenko, Denis	Russia
Kolesnikov, Gennady	Russia
Kopyltsov, Evgeny	Russia
Kovalskiy, Sergey	Russia
Kulikov, Eugeny	Russia
Izmaylov, Adel	Russia
Lykov, Alexander	Russia
Mamonov, Mark	Russia
Mangushlo, Alexander	Russia
Matveev, Vsevolod	Russia
Semenov, Andrey	Russia
Shirshov, Vasily	Russia
Shkolnik, Mikhail	Russia
Skopin, Pavel	Russia
Smirnov, Roman	Russia
Tishchenko, Anton	Russia
Varlamov, Sergey	Russia
Veliev, Eugeny	Russia
Vladimirov, Vladimir	Russia
Garcia Donas, Jesus	Spain
Juarez, Alvaro	Spain
Leivar, Asier	Spain
Llorente, Carlos	Spain
Requena, Ma José	Spain
Rodríguez-Antolín, Alfredo	Spain
Andren, Ove	Sweden
Bjartell, Anders	Sweden
Damber, Jan-Erik	Sweden
Elwaheidy, Amer	Sweden
Haggman, Michael	Sweden
Ljungberg, Borje	Sweden
Norming, Ulf	Sweden
Akdogan, Bulent	Turkey

Arslan, Cagatay	Turkey
Baltaci, Sumer	Turkey
Cal, Cag	Turkey
Cicin, Irfan	Turkey
Ozguroglu, Mustafa	Turkey
Sozen, Sinan	Turkey
Yaman Sezer, Emel	Turkey
Yalcin, Bulent	Turkey
Yigitbasi, Orhan	Turkey
Bondarenko, Igor	Ukraine
Gotsuliak, Iaroslav	Ukraine
Hodos, Valeriy	Ukraine
Hotko, Eugene	Ukraine
Ivashchenko, Petro	Ukraine
Khareba, Gennadii	Ukraine
Kostiuk, Oleksandr	Ukraine
Lychkovskyy, Oleksandr	Ukraine
Lyulko, Oleksiy	Ukraine
Paramonov, Viktor	Ukraine
Pidmurnyak, Oleksiy	Ukraine
Pryanikov, Vadym	Ukraine
Sakalo, Valerii	Ukraine
Sonnyk, Ievgenii	Ukraine
Stakhovsky, Eduard	Ukraine
Stus, Viktor	Ukraine
Aning, Jonathan	United Kingdom
Chowdhury, Simon	United Kingdom
Dixit, Sanjay	United Kingdom
Jones, Rob	United Kingdom
Kumar, Anil	United Kingdom
Leaning, Darren	United Kingdom
Nabi, Ghulam	United Kingdom
Natale, Salvatore	United Kingdom
Protheroe, Andrew	United Kingdom
Sayers, Ian	United Kingdom
Agarwal, Neeraj	United States of America
Agrawal, Manish	United States of America
Babu, Sunil	United States of America
Bailen, James	United States of America
Bardot, Stephen	United States of America
Belkoff, Laurence	United States of America
Bylow, Kathryn	United States of America
Chu, Franklin	United States of America
Cochran, James	United States of America
DeGuenther, Mark	United States of America
Doshi, Gurjyot	United States of America
Dunshee, Curtis	United States of America
Frankel, Jeffrey	United States of America
Gartrell, Benjamin	United States of America

Gervasi, Lawrence	United States of America
Given, Robert	United States of America
Goodman, Michael	United States of America
Hafron, Jason	United States of America
Hauke, Ralph	United States of America
Jalkut, Mark	United States of America
Karsh, Lawrence	United States of America
Lewis, Brian	United States of America
Mannuel, Heather	United States of America
Mehlhoff, Bryan	United States of America
Morris, David	United States of America
Pfanzelter, Nicklas	United States of America
Pieczonka, Christopher	United States of America
Sieber, Paul	United States of America
Saltzstein, Daniel	United States of America
Schiffman, Zvi	United States of America
Sharifi, Roohollah	United States of America
Tutrone, Ronald	United States of America

SUPPLEMENTARY METHODS

Definitions of Secondary End Points

- Time to cytotoxic chemotherapy was defined as the time from randomization to initiation of cytotoxic chemotherapy.
- Time to pain progression was defined as the time from randomization to pain progression (average increase in 2 points from baseline in Brief Pain Inventory-Short Form [BPI-SF] worst pain intensity observed at two consecutive evaluations ≥ 3 weeks apart, with an average worst pain score of >4 in patients who have had no decrease in opioids or initiation of chronic opioids, whichever occurs first).
 - BPI-SF worst pain (item 3) was used for the time to pain progression end point.
 - Scores range from 0 to 10, with lower scores representing lower levels of pain intensity; a change of 2 was the minimally important difference.¹
- Time to chronic opioid use was defined as the time from randomization to chronic opioid use.
 - Chronic opioid use was defined as administration of opioid analgesics for ≥ 3 weeks for oral or ≥ 7 days for nonoral formulations.
 - For patients who were already receiving opioids at study entry, chronic opioid use was defined as a $\geq 30\%$ increase in total daily dose of the opioid analgesics lasting for ≥ 3 weeks for oral or ≥ 7 days for nonoral formulations.
 - Administration of as-needed (eg, not fixed or scheduled dosage) opioid analgesics or extended opioid use for treatment other than the patient's prostate cancer did not require discontinuation from study treatment.
- Time to skeletal-related event was defined as the time from randomization to first observation of a skeletal-related event (symptomatic pathologic fracture, spinal cord compression, radiation to bone, or surgery to bone).

Definitions of Exploratory End Points

- Time to PSA progression was defined as the time from randomization to date of PSA progression based on Prostate Cancer Working Group 2 criteria.²
- Time to second progression-free survival was defined as the time from randomization to first occurrence of investigator-determined disease progression (PSA progression, progression on imaging, or clinical progression) while patient was receiving first subsequent therapy for prostate cancer, or death due to any cause, whichever occurs first.
- Time to symptomatic local progression was defined as the time from randomization to date of symptomatic local progression.

Prostate Cancer Working Group 2 Criteria

Patients in TITAN were assessed for efficacy per modified Response Evaluation Criteria In Solid Tumors version 1.1 and Prostate Cancer Working Group 2 criteria.² Prostate Cancer Working Group 2 criteria for progression are as follows:

- PSA: first $\geq 25\%$ increase from baseline and ≥ 2 ng/mL above the nadir (confirmed by second value 3 or more weeks later)
- Soft-tissue lesions: follow Response Evaluation Criteria in Solid Tumors with caveats
 - Imaging should include a computed tomography scan or magnetic resonance imaging at a minimum; centers with relevant expertise should utilize endorectal magnetic resonance imaging or transrectal ultrasound
 - Only report changes in lymph nodes that were ≥ 2 cm in diameter at baseline
 - Record changes in nodal and visceral soft tissue sites separately
 - Record complete elimination of disease at any site separately
 - Confirm favorable change with a second scan
 - Record changes using a waterfall plot

- Progression at first assessment must be confirmed by a second scan ≥ 6 weeks later
- Bone: appearance of ≥ 2 new lesions, and, for the first reassessment only, a confirmatory scan performed ≥ 6 weeks later indicating a minimum of ≥ 2 new lesions

SUPPLEMENTARY RESULTS

Rash Management

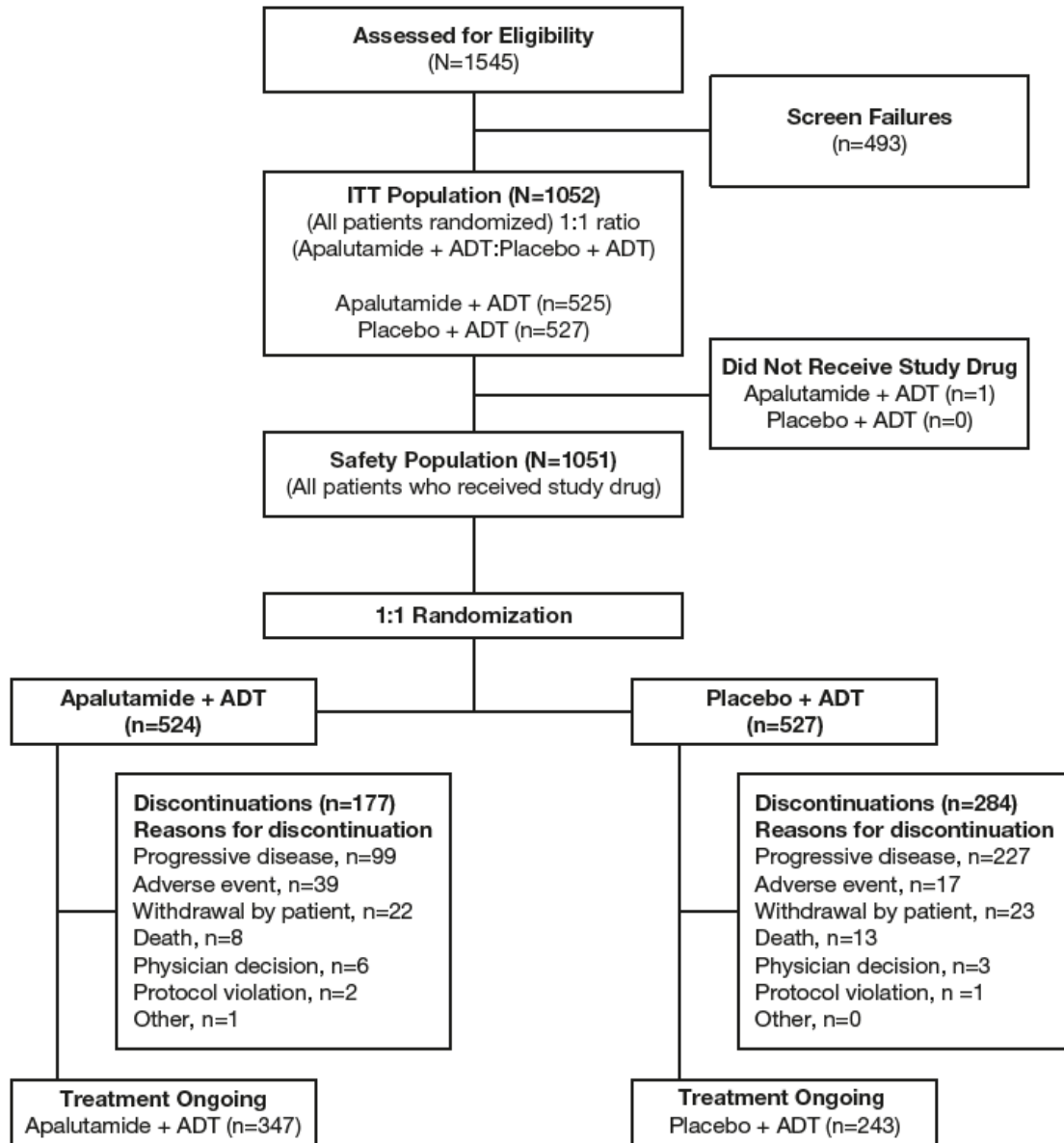
Skin rash associated with apalutamide was commonly described as generalized or maculo-papular.

Treatment-emergent skin rash was reported by 27.1% of patients in the apalutamide group versus 8.5% in the placebo group. Grade 3 rashes were reported with apalutamide (6.3%) and placebo (0.6%) treatment.

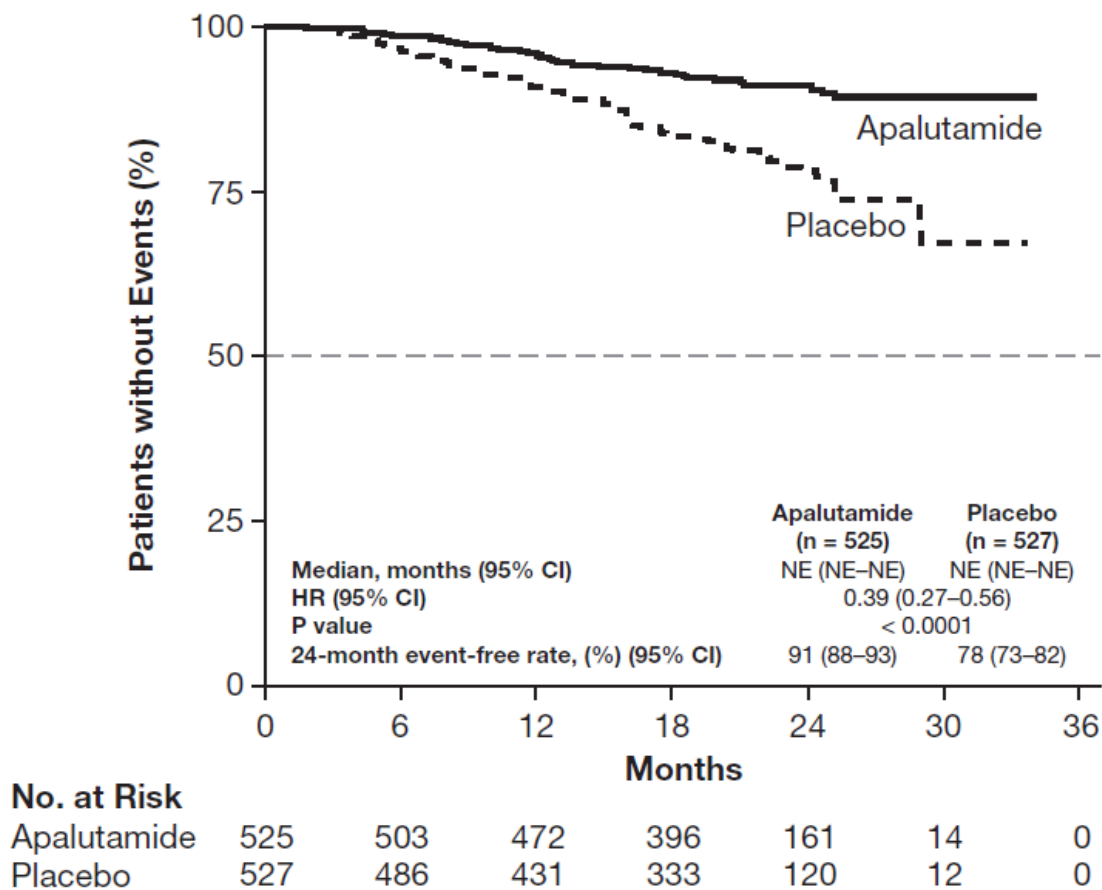
No Stevens-Johnson syndrome or toxic epidermal necrolysis was reported. Skin rash led to treatment discontinuation, dose reduction, and dose interruption in 12 (2.3%), 28 (5.3%), and 44 (8.4%), patients, respectively, in the apalutamide group, and 1 (0.2%), 4 (0.8%), and 5 (0.9%), patients, respectively, in the placebo group. Grade ≥ 3 rash led to discontinuation in 7 (1.3%) and 1 (0.2%) patients in the apalutamide and placebo groups, respectively (Table S4, Supplementary Appendix). For patients who had a skin rash, treatment included topical corticosteroids, oral antihistamines, systemic corticosteroids, drug interruption, and dose reduction. Median time to onset of rash was 81 days in the apalutamide group and 141 days in the placebo group.

SUPPLEMENTARY FIGURES

Supplementary Figure S1. CONSORT Diagram.

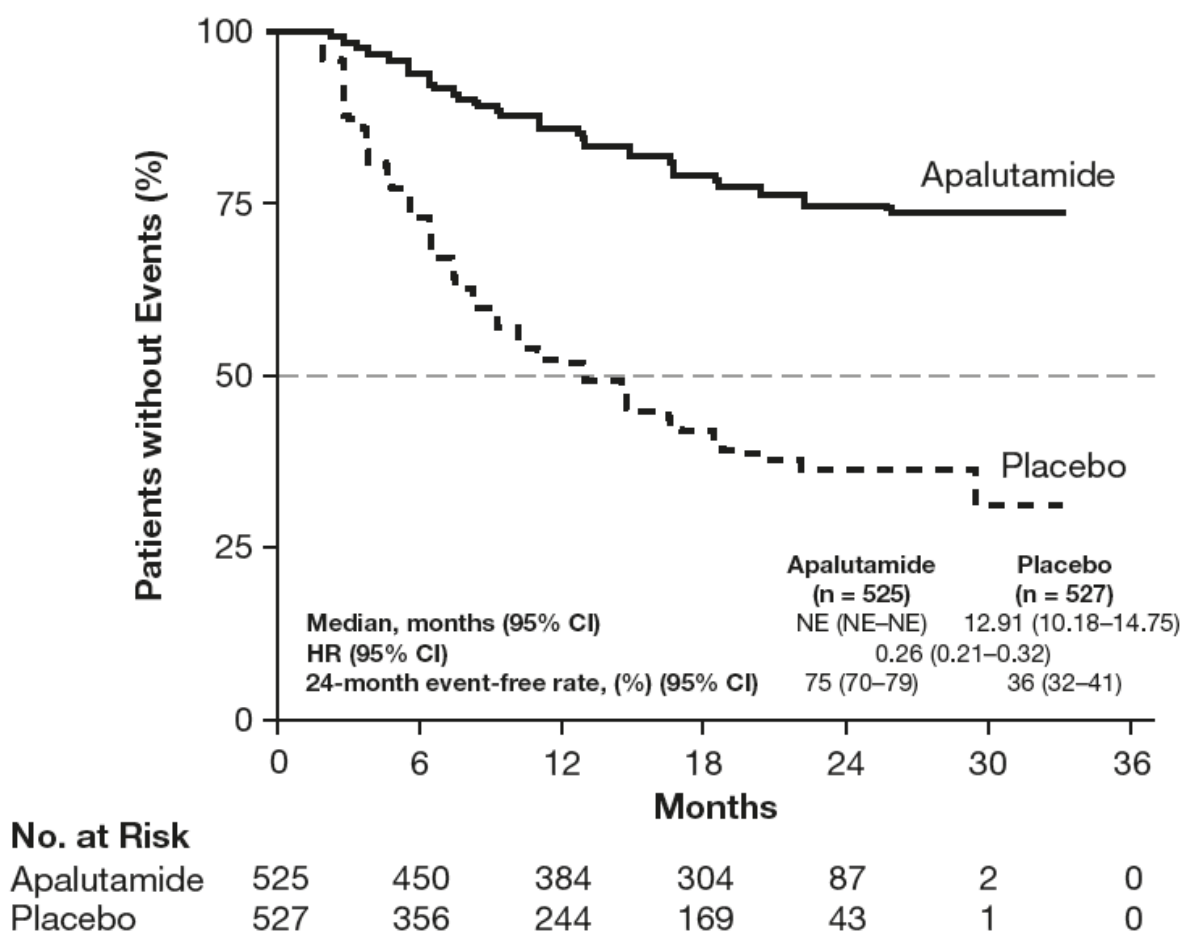


Time to Initiation of Cytotoxic Chemotherapy



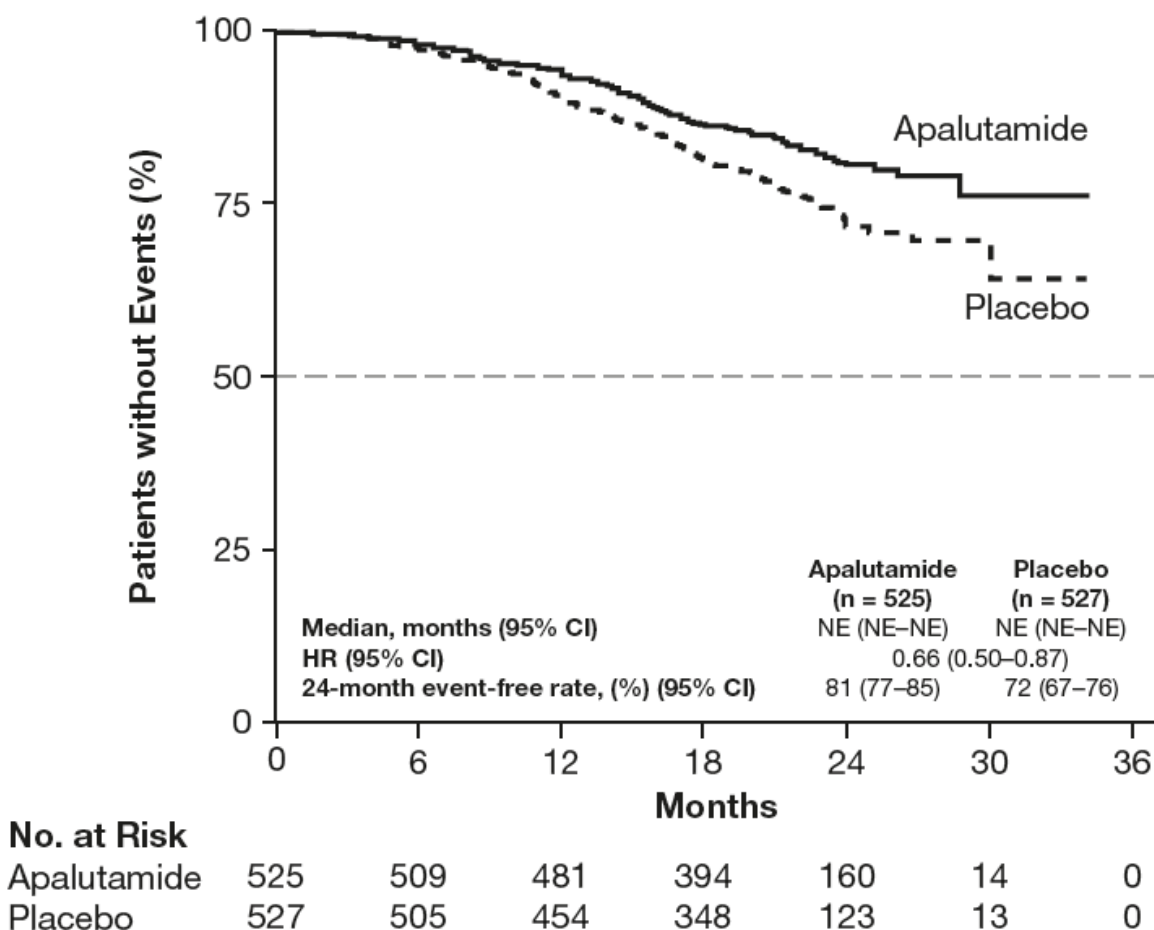
Supplementary Figure S2. **Kaplan-Meier Estimate of Time to Cytotoxic Chemotherapy.** Analyses were performed using a log-rank test with stratification by Gleason score at diagnosis (≤ 7 vs. > 7), region (North America and European Union vs. all other countries), and prior treatment with docetaxel (yes vs. no).

Time to PSA Progression



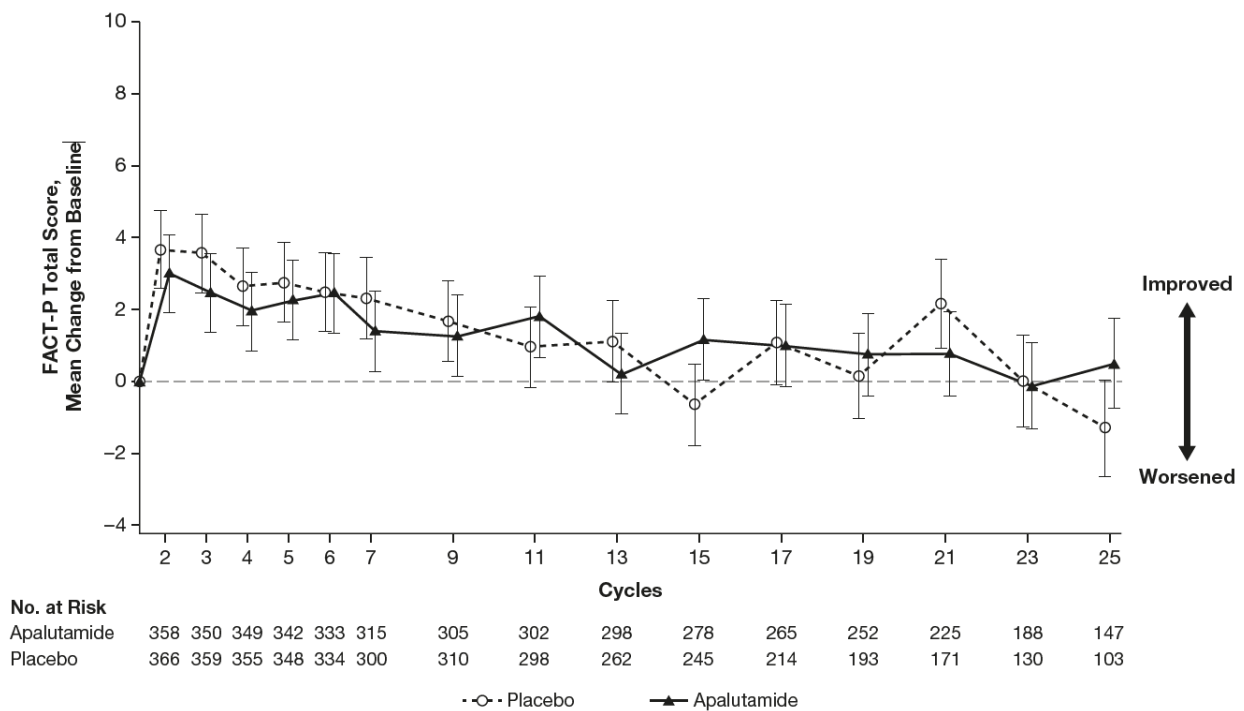
Supplementary Figure S3. **Kaplan-Meier Estimate of Time to PSA Progression.** Time to PSA progression was the time from date of randomization to date of PSA progression, based on Prostate Cancer Working Group 2 criteria. Analyses were performed using a log-rank test with stratification by Gleason score at diagnosis (≤ 7 vs. > 7), region (North America and European Union vs. all other countries), and prior treatment with docetaxel (yes vs. no).

Time to Second Progression-Free Survival



Supplementary Figure S4. **Kaplan-Meier Estimate of Time to Second Progression-free Survival.**

Second progression-free survival was the time from date of randomization to first occurrence of investigator-determined disease progression (PSA progression, progression on imaging, or clinical progression) while patient was receiving first subsequent therapy for prostate cancer or death due to any cause, whichever occurs first. Analyses were performed using a log-rank test with stratification by Gleason score at diagnosis (≤ 7 vs. > 7), region (North America and European Union vs. all other countries), and prior treatment with docetaxel (yes vs. no).



Supplementary Figure S5. **Repeated-Measures Mixed Effects Analyses for Mean Change from Baseline in FACT-P Total Score.** Error bars are standard errors of the mean. Raw FACT-P scores range from 0 to 156, with higher scores indicating more favorable health-related quality of life; a 6- to 10-point change in FACT-P total score would be the minimally important difference. However, this figure presents mean changes in total scores compared with baseline rather than raw total scores. FACT-P, Functional Assessment of Cancer Therapy-Prostate.

Supplementary Table S1. **Demographics and Baseline Disease Characteristics, Expanded List.**

	Apalutamide (n=525)	Placebo (n=527)
Age, median – yr (range)	69 (45–94)	68 (43–90)
Age (yr) – no. (%)		
<65	149 (28.4)	182 (34.5)
65–69	136 (25.9)	108 (20.5)
70–74	107 (20.4)	124 (23.5)
≥75	133 (25.3)	113 (21.4)
Race – no. (%)		
White	354 (67.4)	365 (69.3)
Asian	119 (22.7)	110 (20.9)
Black or African American	10 (1.9)	9 (1.7)
American Indian or Alaska native	6 (1.1)	13 (2.5)
Not reported	11 (2.1)	8 (1.5)
Other	24 (4.6)	22 (4.2)
Multiple	1 (0.2)	0
ECOG PS score – no. (%)	n=525	n=527
0	328 (62.5)	348 (66.0)
1	197 (37.5)	178 (33.8)
2	0	1 (0.2)
Gleason score at initial diagnosis – no. (%)	n=525	n=527
<7	41 (7.8)	39 (7.4)
7	133 (25.3)	130 (24.7)
>7	351 (66.9)	358 (67.9)
Tumor, lymph node, metastatic stage at initial diagnosis – no. (%)	n=525	n=527
T0	1 (0.2)	0
T1	41 (7.8)	27 (5.1)
T2	146 (27.8)	110 (20.9)
T3	210 (40.0)	225 (42.7)
T4	76 (14.5)	105 (19.9)
TX	51 (9.7)	60 (11.4)
N0	212 (40.4)	216 (41.0)
N1	199 (37.9)	184 (34.9)
NX	114 (21.7)	127 (24.1)
M0	85 (16.2)	59 (11.2)
M1	411 (78.3)	441 (83.7)
MX	29 (5.5)	27 (5.1)
Disease volume – no. (%)		
Low	200 (38.1)	192 (36.4)
High	325 (61.9)	335 (63.6)
Extent of disease at study entry – no. (%)		
Bone	525 (100.0)	527 (100.0)
Bone only	289 (55.0)	269 (51.0)
Lymph node	199 (37.9)	219 (41.6)
Visceral and bone	56 (10.7)	72 (13.7)
Lung	47 (9.0)	64 (12.1)
Liver	12 (2.3)	13 (2.5)

Soft tissue and bone	22 (4.2)	27 (5.1)
Median time from initial diagnosis to randomization (range) – mo	4.1 (0.5–222.9)	4.0 (0.7–341.4)
Prior docetaxel – no. (%)		
Yes*	58 (11.0)	55 (10.4)
No	467 (89.0)	472 (89.6)
Median no. of cycles of docetaxel among patients with prior docetaxel	6	6
Prior therapy for localized prostate cancer – no. (%)		
Prostatectomy or radiotherapy	94 (17.9)	79 (15.0)
Prostatectomy only	26 (5.0)	27 (5.1)
Radiotherapy only	47 (9.0)	39 (7.4)
Both prostatectomy and radiotherapy	21 (4.0)	13 (2.5)
Median PSA (range) – µg/L	5.97 (0–2682)	4.02 (0–2229)
Median LDH (range) – U/L	178 (88–1248)	179 (85–1514)
Median alkaline phosphatase (range) – U/L	98 (26–5193)	94 (32–3892)
Mean baseline BPI-SF pain score† – no. (%)	198 (37.7)	200 (38.0)
0 (no pain)	195 (37.1)	207 (39.3)
1 to 3 (mild pain)	98 (18.7)	95 (18.0)
4 to 7 (moderate pain)	12 (2.3)	11 (2.1)
8 to 10 (severe pain)		
Median baseline FACT-P total score‡	113	113

ECOG PS, Eastern Cooperative Oncology Group performance status; FACT-P, Functional Assessment of Cancer Therapy-Prostate; PSA, prostate-specific antigen.

* 27 patients (46.6%) in the apalutamide group and 22 patients (40.0%) in the placebo group were N1 at diagnosis.

† Scores range 0 to 10, with lower scores representing lower levels of pain intensity; a change of 2 was the minimally important difference.¹

‡ Scores range from 0 to 156, with higher scores indicating more favorable health-related quality of life. A 6- to 10-point change in FACT-P total score is the minimal important difference.

Supplementary Table S2. **Prior Prostate Cancer Therapy.**

	Apalutamide	Placebo
	(n=525)	(n=527)
Prostatectomy or radiotherapy – no. (%)	94 (17.9)	79 (15.0)
Prostatectomy only	26 (5.0)	27 (5.1)
Radiotherapy only	47 (9.0)	39 (7.4)
Both prostatectomy and radiotherapy	21 (4.0)	13 (2.5)
Hormonal therapy – no. (%)	525 (100)	527 (100)
First-generation antiandrogen	352 (67.0)	361 (68.5)
Gonadotropin-releasing hormone agonist	462 (88.2)	455 (86.3)
Gonadotropin-releasing hormone antagonist	56 (10.7)	53 (10.1)
Bilateral orchiectomy	33 (6.3)	40 (7.6)
Docetaxel – no. (%)	58 (11.0)	55 (10.4)
Vandetanib – no. (%)	1 (0.2)	0

Supplementary Table S3. **First Subsequent Systemic Prostate Cancer Therapy.**

	Apalutamide (n=525)	Placebo (n=527)
Patients who discontinued treatment for any reason* and remained alive (<i>denominator for first subsequent therapy calculations below</i>) – no.	170	271
Patients who discontinued because of disease progression – no. (%)	99 (18.9)	227 (43.1)
Patients with adverse events leading to discontinuation – no. (%)	42 (8.0)	28 (5.3)
Patients with life-prolonging subsequent therapy for prostate cancer† – no. (%)	64 (37.6)	165 (60.9)
Patients receiving subsequent systemic prostate cancer therapy at data cutoff‡ – no. (%)	87 (51.2)	190 (70.1)
First subsequent therapy		
Hormonal therapy – no. (%)	44 (25.9)	98 (36.2)
Abiraterone acetate plus prednisone	21 (12.4)	45 (16.6)
Bicalutamide	16 (9.4)	31 (11.4)
Enzalutamide	3 (1.8)	17 (6.3)
Other§	4 (2.4)	5 (1.8)
Chemotherapy – no. (%)	35 (20.6)	73 (26.9)
Docetaxel	29 (17.1)	67 (24.7)
Cabazitaxel	1 (0.6)	2 (0.7)
Other**	5 (2.9)	4 (1.5)

Other therapy – no. (%)	8 (4.7)	19 (7.0)
Radium-223	2 (1.2)	4 (1.5)
Sipuleucel-T	2 (1.2)	5 (1.8)
Other***	4 (2.4)	10 (3.7)

* Possible reasons for discontinuation were disease progression, adverse event, withdrawal by patient, death, physician decision, and protocol violation.

† Included docetaxel, abiraterone acetate plus prednisone, enzalutamide, cabazitaxel, radium-223, and sipuleucel-T.

‡ Some patients were unblinded after discontinuation and prior to first subsequent therapy to allow enrollment in a subsequent clinical trial.

§ Included diethylstilbestrol, flutamide, and cyproterone.

** Included etoposide, paclitaxel, estramustine, carboplatin, and cisplatin.

*** Included zoledronic acid, clodronate, prednisolone, and prednisone.

Supplementary Table S4. **Treatment-Emergent Adverse Events Leading to Treatment**

Discontinuation, Dose Reduction, and Dose Interruption.

	Apalutamide		Placebo	
	(n=524)		(n=527)	
	All Grades	Grade ≥3	All Grades	Grade ≥3
Discontinuation				
Patients with treatment-emergent adverse events leading to discontinuation – no. (%)	42 (8.0)	26 (5.0)	28 (5.3)	22 (4.2)
Rash*	12 (2.3)	7 (1.3)	1 (0.2)	1 (0.2)
New cancer neoplasm†	7 (1.3)	5 (1.0)	5 (0.9)	3 (0.6)
Infections‡	1 (0.2)	1 (0.2)	3 (0.6)	0
Fatigue	4 (0.8)	1 (0.2)	0	0
Ischemic cardiac events§	2 (0.4)	2 (0.4)	2 (0.4)	2 (0.4)
Seizure	2 (0.4)	1 (0.2)	1 (0.2)	0
Pulmonary embolism	1 (0.2)	1 (0.2)	2 (0.4)	1 (0.2)
Sudden death	0	0	2 (0.4)	2 (0.4)
Cerebrovascular disorder	1 (0.2)	1 (0.2)	0	0
Parkinson's disease	1 (0.2)	1 (0.2)	0	0
Radiculopathy	1 (0.2)	1 (0.2)	0	0
Cognitive disorder	0	0	1 (0.2)	0

Intracranial hemorrhage	0	0	1 (0.2)	1 (0.2)
Loss of consciousness	0	0	1 (0.2)	1 (0.2)
Atrial fibrillation	1 (0.2)	1 (0.2)	0	0
Cardio-respiratory arrest	1 (0.2)	1 (0.2)	0	0
Cardiogenic shock	1 (0.2)	1 (0.2)	0	0
Cardiac failure	0	0	1 (0.2)	1 (0.2)
Sudden cardiac death	1 (0.2)	1 (0.2)	0	0
Hypothermia	0	0	1 (0.2)	1 (0.2)
Anxiety	1 (0.2)	0	0	0
Euphoric mood	1 (0.2)	0	0	0
Suicide	0	0	1 (0.2)	1 (0.2)
Large intestinal ulcer perforation	1 (0.2)	1 (0.2)	0	0
Lip swelling	1 (0.2)	0	0	0
Dyspnea	0	0	1 (0.2)	0
Pleural effusion	0	0	1 (0.2)	1 (0.2)
Pulmonary edema	0	0	1 (0.2)	1 (0.2)
Respiratory failure	0	0	1 (0.2)	1 (0.2)
Dehydration	1 (0.2)	0	0	0

Hot flush	1 (0.2)	0	0	0
Subdural hemorrhage	0	0	1 (0.2)	1 (0.2)
Alanine aminotransferase increased	0	0	1 (0.2)	1 (0.2)
	Dose Reduction			
Patients with treatment-emergent adverse events leading to dose reduction – no. (%)	37 (7.1)	19 (3.6)	11 (2.1)	1 (0.2)
Rash*	28 (5.3)	11 (2.1)	4 (0.8)	1 (0.2)
Aspartate aminotransferase or alanine aminotransferase increased	0	0	4 (0.8)	0
Fatigue	2 (0.4)	2 (0.4)	0	0
Headache	1 (0.2)	0	1 (0.2)	0
Hypertension	1 (0.2)	1 (0.2)	1 (0.2)	0
Neutropenia	2 (0.4)	2 (0.4)	0	0
Cognitive disorder	1 (0.2)	1 (0.2)	0	0
General physical health deterioration	1 (0.2)	1 (0.2)	0	0
Hot flush	1 (0.2)	0	0	0
Asthenia	0	0	1 (0.2)	0
Dehydration	1 (0.2)	0	0	0
Lethargy	0	0	1 (0.2)	0

Pain**	1 (0.2)	0	0	0
Performance status decreased	0	0	1 (0.2)	0
Vertigo	1 (0.2)	1 (0.2)	0	0
Weight decreased	1 (0.2)	1 (0.2)	0	0
	Dose Interruption			
Patients with treatment-emergent events leading to dose interruption – no. (%)	104 (19.8)	68 (13.0)	63 (12.0)	43 (8.2)
Rash*	44 (8.4)	20 (3.8)	5 (0.9)	2 (0.4)
Aspartate aminotransferase or alanine aminotransferase increased	7 (1.3)	4 (0.8)	12 (2.3)	7 (1.3)
Pain**	9 (1.7)	4 (0.8)	7 (1.3)	3 (0.6)
Infections‡	6 (1.1)	4 (0.8)	9 (1.7)	3 (0.6)
Hypertension	6 (1.1)	6 (1.1)	6 (1.1)	6 (1.1)
Fatigue	6 (1.1)	1 (0.2)	2 (0.4)	1 (0.2)
Vomiting	3 (0.6)	0	4 (0.8)	0
Anemia	2 (0.4)	0	3 (0.6)	2 (0.4)
Asthenia	3 (0.6)	3 (0.6)	2 (0.4)	1 (0.2)
Decreased appetite	2 (0.4)	1 (0.2)	3 (0.6)	1 (0.2)
New cancer neoplasm†	3 (0.6)	1 (0.2)	1 (0.2)	1 (0.2)

Diarrhea	1 (0.2)	0	3 (0.6)	0
Chronic obstructive pulmonary disease	2 (0.4)	2 (0.4)	2 (0.4)	2 (0.4)
Pyrexia	3 (0.6)	0	1 (0.2)	0
Urinary retention	2 (0.4)	2 (0.4)	2 (0.4)	2 (0.4)
Arthralgia	2 (0.4)	0	1 (0.2)	0
Insomnia	3 (0.6)	0	0	0
Acute kidney injury	1 (0.2)	1 (0.2)	1 (0.2)	1 (0.2)
Anxiety	2 (0.4)	0	0	0
Cerebrovascular accident	1 (0.2)	1 (0.2)	1 (0.2)	0
Dizziness	1 (0.2)	0	1 (0.2)	0
Dyspnea	1 (0.2)	0	1 (0.2)	1 (0.2)
Fall	1 (0.2)	1 (0.2)	1 (0.2)	1 (0.2)
Headache	1 (0.2)	0	1 (0.2)	1 (0.2)
Hematuria	2 (0.4)	2 (0.4)	0	0
Hot flush	1 (0.2)	0	1 (0.2)	0
Hypersensitivity	1 (0.2)	0	1 (0.2)	0
Hypertriglyceridemia	2 (0.4)	2 (0.4)	0	0
Hyponatremia	1 (0.2)	1 (0.2)	1 (0.2)	1 (0.2)

Muscular weakness	1 (0.2)	0	1 (0.2)	1 (0.2)
Nausea	1 (0.2)	0	1 (0.2)	0
Neutropenia	2 (0.4)	2 (0.4)	0	0
Spinal cord compression	1 (0.2)	1 (0.2)	1 (0.2)	1 (0.2)
Acute respiratory failure	1 (0.2)	1 (0.2)	0	0
Allergy to chemicals	1 (0.2)	1 (0.2)	0	0
Angina unstable	0	0	1 (0.2)	1 (0.2)
Atrial fibrillation	1 (0.2)	1 (0.2)	0	0
Bile duct obstruction	0	0	1 (0.2)	1 (0.2)
Blood alkaline phosphatase increased	1 (0.2)	1 (0.2)	0	0
Cardiac amyloidosis	0	0	1 (0.2)	1 (0.2)
Cardiac disorder	0	0	1 (0.2)	1 (0.2)
Cardiac failure	0	0	1 (0.2)	0
Cardiac failure congestive	1 (0.2)	1 (0.2)	0	0
Cheilitis	1 (0.2)	0	0	0
Chronic hepatitis	1 (0.2)	0	0	0
Cognitive disorder	1 (0.2)	1 (0.2)	0	0
Confusional state	1 (0.2)	1 (0.2)	0	0

Constipation	1 (0.2)	0	0	0
Dehydration	0	0	1 (0.2)	0
Depression	1 (0.2)	0	0	0
Diabetes mellitus	1 (0.2)	1 (0.2)	0	0
Diplegia	0	0	1 (0.2)	1 (0.2)
Dysuria	1 (0.2)	1 (0.2)	0	0
Edema	1 (0.2)	0	0	0
Eyelid edema	1 (0.2)	0	0	0
Facial edema	1 (0.2)	0	0	0
Febrile neutropenia	1 (0.2)	1 (0.2)	0	0
Gamma-glutamyltransferase increased	1 (0.2)	1 (0.2)	0	0
Gastric ulcer perforation	1 (0.2)	1 (0.2)	0	0
Gastritis	0	0	1 (0.2)	1 (0.2)
General physical health deterioration	1 (0.2)	1 (0.2)	0	0
Hallucination	1 (0.2)	0	0	0
Hematoma	1 (0.2)	1 (0.2)	0	0
Hemorrhagic erosive gastritis	1 (0.2)	1 (0.2)	0	0
Hemorrhage intracranial	1 (0.2)	1 (0.2)	0	0

Hepatic cirrhosis	1 (0.2)	0	0	0
Hepatic failure	0	0	1 (0.2)	1 (0.2)
Hip fracture	1 (0.2)	1 (0.2)	0	0
Hypokalemia	1 (0.2)	1 (0.2)	0	0
Hydronephrosis	0	0	1 (0.2)	1 (0.2)
Hyperbilirubinemia	1 (0.2)	1 (0.2)	0	0
Influenza-like illness	1 (0.2)	0	0	0
Large intestinal ulcer perforation	1 (0.2)	1 (0.2)	0	0
Mental status changes	1 (0.2)	0	0	0
Musculoskeletal stiffness	0	0	1 (0.2)	0
Ischemic cardiac events§	1 (0.2)	1 (0.2)	0	0
Osteoarthritis	0	0	1 (0.2)	1 (0.2)
Osteoporosis	1 (0.2)	0	0	0
Pathological fracture	0	0	1 (0.2)	1 (0.2)
Peripheral edema	1 (0.2)	0	0	0
Peripheral swelling	0	0	1 (0.2)	1 (0.2)
Proctalgia	1 (0.2)	0	0	0
Renal injury	0	0	1 (0.2)	0

Rheumatoid arthritis	1 (0.2)	1 (0.2)	0	0
Seizure	1 (0.2)	0	0	0
Stomatitis	1 (0.2)	0	0	0
Subarachnoid hematoma	1 (0.2)	1 (0.2)	0	0
Subarachnoid hemorrhage	1 (0.2)	0	0	0
Subdural hemorrhage	0	0	1 (0.2)	1 (0.2)
Syncope	1 (0.2)	1 (0.2)	0	0
Thrombocytopenia	1 (0.2)	0	0	0
Tongue ulceration	1 (0.2)	0	0	0
Urinary incontinence	1 (0.2)	0	0	0
Urinary tract obstruction	1 (0.2)	1 (0.2)	0	0
Urethral stenosis	0	0	1 (0.2)	0
Vertigo	1 (0.2)	1 (0.2)	0	0
Weight increased	0	0	1 (0.2)	0

* Rash was a grouped term including eczema, eczema nummular, exfoliative rash, generalized rash, rash, dermatitis, dermatitis exfoliative, psoriasis, hyperhidrosis, maculo-papular rash, macular rash, drug eruption, erythema multiforme, lichenoid keratosis, pruritus, pruritic rash, skin exfoliation, skin lesion, skin reaction, and toxic skin eruption.

† New cancer neoplasms (benign, malignant, and unspecified [including cysts and polyps]) was a grouped term including malignant lung neoplasms, colon adenoma, colon neoplasms, leiomyosarcoma,

oropharyngeal cancer, papilloma, rectal adenocarcinoma, small-cell carcinoma, gastric adenocarcinoma, colon adenocarcinoma, bladder cancer, and non-small-cell lung cancer.

‡ Infections was a grouped term including urinary tract infection, *Klebsiella* infection, sepsis, urosepsis, fungal infection, localized infection, lung infection, pneumonia, fungal respiratory tract infection, acarodermatitis, cellulitis, bacterial bronchitis, influenza, necrotizing fasciitis, and viral upper respiratory tract infection.

§ Ischemic cardiac events was a grouped term including myocardial infarction, acute myocardial infarction, acute coronary syndrome.

** Pain was a grouped term including pain in extremity, back pain, bone pain, upper abdominal pain, lower abdominal pain, musculoskeletal pain, neck pain, and noncardiac chest pain.

Supplementary Table S5. **Investigator-Reported Cause of Death on Study in the Safety Population.**

	Apalutamide	Placebo
	(n=524)	(n=527)
All deaths within 30 days of last dose – no. (%)	18 (3.4)	23 (4.4)
Death due to prostate cancer	8 (1.5)	7 (1.3)
Death due to adverse event	10 (1.9)	16 (3.0)
Adverse events leading to death – no. (%)		
Respiratory failure	1 (0.2)	2 (0.4)
Acute kidney injury	2 (0.4)	0
Acute myocardial infarction	1 (0.2)	1 (0.2)
Sudden death	0	2 (0.4)
Acute coronary syndrome	0	1 (0.2)
Cardiac failure	0	1 (0.2)
Cardio-respiratory arrest	1 (0.2)	0
Cardiogenic shock	1 (0.2)	0
Cerebrovascular accident	1 (0.2)	0
Death	0	1 (0.2)
Hypothermia	0	1 (0.2)
Intracranial hemorrhage	0	1 (0.2)

Large intestinal ulcer perforation	1 (0.2)	0
Myocardial infarction	1 (0.2)	0
Pulmonary embolism	0	1 (0.2)
Sepsis	0	1 (0.2)
Subdural hemorrhage	0	1 (0.2)
Sudden cardiac death	1 (0.2)	0
Suicide	0	1 (0.2)
Urosepsis	0	1 (0.2)
Vascular rupture	0	1 (0.2)

Supplementary Table S6. **Estimates of Subdistribution Hazard Ratio of the Apalutamide Group Relative to the Placebo Group.** A post-hoc analysis accounting for the competing risk of death was performed based on the Fine and Gray model.³ The estimates of a subdistribution hazard ratio of the apalutamide group relative to the placebo group, along with the respective 95% confidence limits and the P values from the Wald Chi-Square test, are presented. The results of this post-hoc analysis supported and confirmed the preplanned analysis results that are currently presented in the manuscript.

			95% Confidence Limits of the	
	P Value	Hazard Ratio	Hazard Ratio	
Time to cytotoxic chemotherapy	<0.0001	0.408	0.286	0.581
Time to pain progression	0.1540	0.843	0.667	1.066
Time to chronic opioid use	0.2004	0.789	0.549	1.134
Time to skeletal-related event	0.2759	0.818	0.570	1.174

SUPPLEMENTARY REFERENCES

1. Mathias SD, Crosby RD, Qian Y, Jiang Q, Dansey R, Chung K. Estimating minimally important differences for the worst pain rating of the Brief Pain Inventory-Short Form. *J Support Oncol* 2011;9:72-8.
2. Scher HI, Halabi S, Tannock I, et al. Design and end points of clinical trials for patients with progressive prostate cancer and castrate levels of testosterone: recommendations of the Prostate Cancer Clinical Trials Working Group. *J Clin Oncol* 2008;26:1148-59.
3. Fine JP, Gray RJ. A proportional hazards model for the subdistribution of a competing risk. *J Am Stat Assoc* 1999;94:496-509.